

SHREENAND SACHIN BHATTAD

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EDUCATION

B.Sc (Hons) Computer Science · Minor in Economics

2023 - 2027

Ashoka University · Relevant Coursework: Algorithms & Data Structures, Operating Systems, Machine Learning, Computer Vision, Graph Theory & Network Science, Information Security, Accounting & Finance, Probability & Statistics

Class X, Symbiosis School Nashik · CBSE

95% | 2021

INTERNSHIP EXPERIENCE

Eka.care

May 2026 - Present

AI Research & Engineering Intern · India's leading healthcare digitisation platform - AI tools for 100k+ doctors and patients across clinical speech, records, and real-world intelligence

- **Mining 1M+ structured EHR records** using SQL and scalable Pandas pipelines; applying unsupervised clustering (DBSCAN, hierarchical agglomerative) to identify seasonal disease clusters, drug co-prescription networks, and regional diagnosis gaps across patient cohorts and geographies.
- **Fine-tuning ClinicalBERT** for multi-label ICD-10 diagnosis classification over free-text clinical notes; implementing **temperature-scaled confidence calibration** to flag low-certainty model outputs for clinician review and active-learning re-annotation.
- **Building LSTM and Transformer sequence models** over longitudinal patient records to surface temporal patterns in referral chains and chronic disease progression; generating **population-level risk stratification scores** for public health policy briefs.
- **Applying chi-square, Mann-Whitney U, multivariate logistic regression, and Kaplan-Meier survival analysis** to validate statistical significance of observed clinical patterns; structuring findings for academic publication on India's primary care landscape.

ARTPARK @ IISc

May 2026 - Present

GenAI DevOps Intern · India's leading deep-tech AI and Robotics incubator, established by IISc Bengaluru with support from DST, Govt. of India

- **Architecting CI/CD pipelines** via GitHub Actions for a GenAI web application: automated test gates, Docker image builds, and zero-downtime deployments to GPU-backed Linux servers, cutting release cycle time significantly.
- **Configuring production-grade Linux infrastructure** (Ubuntu/Debian): nginx reverse proxy, systemd service orchestration, SSL/TLS management, SSH hardening, and **IAM role policies on AWS (EC2, S3, CloudWatch)** for a multi-service LLM stack.
- **Centralising observability** using Grafana/Loki for log aggregation and dashboarding; alerting on p95 latency spikes, disk saturation, and service health with **automated failure notifications** across distributed services.
- **Benchmarking LLM serving frameworks** (Ollama, vLLM) on accuracy-latency-cost trade-offs; evaluating quantized and fine-tuned model variants to establish performance baselines for domain-specific GenAI queries over large public datasets.
- **Building Docker Compose staging environments** for pre-production validation; implementing automated **AWS S3 backup workflows** and cloud storage sync to enforce data integrity and disaster-recovery guarantees.

New and Modern

Dec 2025 - Apr 2026

Software Engineering Intern · Tech-powered luxury PropTech brokerage in NYC: new construction, premarket sourcing, and data-driven real estate at scale

- **Owned full-stack backend features** in Next.js and PostgreSQL across core product flows, improving page render time by **~30%** and p99 query latency by **~25%** via composite indexing, query plan analysis, schema normalization, and join rewrites serving thousands of concurrent property transactions.
- **Designed REST APIs** with typed contracts and predictable data shapes for **high-concurrency access**; applied rate limiting, connection pooling, and backpressure controls to sustain throughput under demand spikes across a distributed agent network.
- **Built production-grade Python ETL pipelines** with full test coverage, structured logging, schema-drift handling, and **idempotent incremental loads**, ingesting multi-source data into a normalized PostgreSQL schema for analytics workflows.
- **Improved AWS/GCP observability** via structured log instrumentation, metric dashboards, and data-quality gates; introduced **Redis caching** to reduce downstream DB load by an estimated **~40%** on hot read paths.

PsyFi Labs

Aug 2025 - Present

Product Engineering Intern · Behavioral science consultancy helping startups optimize product adoption through data-driven insights

- **Led full-stack development** of an automated behavioral-insights dashboard; trained and **productionized an ML user-profiling model** with calibrated confidence scores, integrating async inference into the backend for real-time profile updates with sub-200ms response SLA.
- **Engineered multi-source ETL pipelines** using Google Sheets Apps Script, n8n, and the OpenAI API; designed **incremental load logic with schema validation** to ensure idempotency across concurrent data sources and eliminate manual reporting overhead.
- **Automated pipeline orchestration and model-performance tracking**; translated ML outputs into product requirements and owned Webflow frontend data-binding and visualization logic.

Koita Centre for Digital Health, Ashoka (KCDH-A)

Jan 2025 - Aug 2025

Data Insights Research Intern · Interdisciplinary research centre advancing AI/ML for healthcare, genomics, and India's digital public health ecosystem

- Analyzed **accelerometer and whole-genome data from 5,000+ biobank participants** using Python and R; applied mixed-effects regression and polygenic score analysis to quantify gene-environment interactions on cardiorespiratory outcomes.

- **Built parallel batch pipelines** to preprocess **2+ TB of wearable sensor data** via multiprocessing and chunked I/O, reducing wall-clock preprocessing time from hours to minutes.
- **Engineered time-series features** (MFCC, FFT spectral bands, DTW distance matrices) and trained CNN classifiers; generated **probabilistic cohort-level cardiorespiratory risk scores** achieving publication-grade accuracy on held-out evaluation cohorts.

TECHNICAL PROJECTS

Multi-Task Visual Recognition & Novel View Synthesis - Computer Vision / Deep Learning

Mar - May 2026

- **Fine-tuned ConvNeXt-Base (88M parameters)** on a **131k-image 80-class benchmark**: two-stage progressive resolution curriculum (224 to 320px), layer-wise LR decay, stochastic depth, MixUp/CutMix augmentation, achieving **95.87% Top-1 accuracy** (+4.87pp over baseline).
- **Zero-shot instance segmentation** via SAM ViT-H with open-vocabulary CLIP classification; perspective homography (4-point DLT) for BEV correction increased mask yield **2.2x**; coarse-to-fine NeRF reconstruction achieved **22.00 dB PSNR** on the NVIDIA evaluation set.

Deepvue Bank Statement Intelligence - Applied NLP / Financial Data Infrastructure

Nov 2025 - May 2026

Built for Deepvue.ai - unified customer insights and fraud-decisioning APIs for fintechs and NBFCs.

- **Designed a multi-stage transaction classification pipeline** over **100k+ bank narrations**: rule-based FSM for deterministic labels, inverted-index codebook for exact lookup, and char/word n-gram TF-IDF into calibrated LogisticRegression with **confidence-gated fallback**, achieving high-precision classification across heterogeneous narration formats.
- **Implemented UPI counterparty disambiguation** using regex tokenization and schema-enforced taxonomy resolution; built post-salary **behavioral analytics across 12 spend categories** with savings/expenditure ratio time series for financial health scoring.

Cloud-Based Crowdsourced Issue Tracker - Full-Stack & Distributed Infrastructure

Oct - Dec 2025

- **Built an end-to-end geo-spatial reporting system** with geo-tagged pins, live status, comments, and image uploads; Dockerized backend on AWS EC2 behind Nginx with PostgreSQL + PostGIS for spatial queries and S3 with signed URLs for media storage.
- **Implemented JWT auth, RBAC, rate limiting, and Redis caching** for hot geo-lookups; vector-tile map rendering via Leaflet sustaining low-latency responses under concurrent user load.

Forest Fire Impact & Recovery Analysis - Geospatial Data Science / Remote Sensing

Apr - Nov 2025

- **Mapped post-fire landcover change in the Himalayas** using Landsat-8 and Sentinel-2 in Google Earth Engine; computed **NDVI and dNBR time series** to quantify burn extent and track vegetation recovery at 10m spatial resolution across 2,000+ km² of terrain.
- **Built automated Python preprocessing pipelines** using rasterio, GDAL, and geopandas; trained a scikit-learn **Random Forest classifier** to label burned vs. recovering pixels and forecast recovery trajectories.

POSITIONS OF RESPONSIBILITY

Director, Research & Data Insight · Ashoka Data Society

Jun 2025 - Present

Leading interdisciplinary data projects with external partners; delivering reports, dashboards, and technical newsletter summaries for a 200+ member society.

Head of Technical Operations · Ashoka Economics Society

Sep 2024 - Present

Architecting and managing web platforms and digital infrastructure for the largest academic society at Ashoka University (~1,000+ members).

Growth Department · IEEE Ashoka Student Chapter

Jan 2024 - Present

Networked with students and professionals across 10+ universities; co-organized IEEE technical workshops and events.

CERTIFICATIONS & EXTRACURRICULAR

Harvard CS50x (Harvard University) · Machine Learning (Columbia University) · Responsive Web Design (freeCodeCamp) · Entrepreneurship Modules (EDI India)

Badminton: State & District level competitor, School Team Captain, Ashoka University Badminton Team (2013 - 2025)